

Optimizing Low-Resource Language Models via Differentiated Token-Masking: A Framework for Self-Curated Difficulty Alignment

Introduction: The Low-Resource Curation Paradigm

The trajectory of large language model (LLM) research has increasingly prioritized data efficiency and targeted optimization over raw parameter scaling.¹ As models approach the asymptotic limits of available human-generated text, the focus has shifted toward self-curated dataset optimization, wherein models actively filter, refine, and select their own training data to maximize supervised fine-tuning (SFT) efficacy.² This paradigm is particularly critical for lightweight architectures, where parameter capacity acts as a strict bottleneck. In such constrained environments, exposing a model to overly complex, noisy, or dense data does not simply result in inefficient learning; it actively degrades alignment and induces catastrophic forgetting by exceeding the model's representational capacity.⁴ Consequently, the ability to accurately assess and select training data based on its intrinsic cognitive difficulty has emerged as a paramount challenge in contemporary machine learning.²

This report details an exhaustive theoretical, mathematical, and experimental framework for investigating self-curated dataset optimization through the precise lens of token-masking mechanics. The focal point of this investigation utilizes the Qwen-2.5-0.5B model, an advanced lightweight architecture specifically chosen for its low hardware requirements, rapid iteration capabilities, and robust baseline performance across diverse cognitive tasks.⁵ By establishing two independent data tracks—one dedicated to semantic understanding and linguistic nuance, and the other rigorously focused on logical and mathematical reasoning—the research isolates the impact of specialized data curation on distinct cognitive modalities.⁷

The core theoretical novelty of this framework lies in the comparative analysis of two distinct token-masking strategies used to assign a "difficulty score" to candidate training samples dynamically. Strategy A, conceptualized as Input Masking, obscures key informational tokens within the prompt context to evaluate the model's generation stability and its reliance on specific instructional cues. Strategy B, defined as Output Masking, obscures target tokens within the historical answer sequence to evaluate the model's predictive certainty and internal logical consistency. By defining a mathematically rigorous yet computationally lightweight framework for these difficulty scores, the research enables the selective curation of fine-tuning subsets that perfectly match the model's optimal learning threshold.⁴

This comprehensive proposal encompasses a deep literature review to contextualize the prevailing research gap, formal mathematical definitions for the proposed difficulty metrics grounded in information theory, a highly detailed experimental protocol tailored for single-GPU consumer environments, a robust multi-dimensional evaluation matrix, and a strategic

framework designed to ensure the academic validity of the findings irrespective of the empirical outcomes.

Comprehensive Literature Review and Contextualization

The foundation of this research rests upon three intersecting domains of machine learning: self-distillation and automated data selection, curriculum learning guided by perplexity and difficulty metrics, and token-level masking mechanisms. Analyzing the current state of these fields reveals critical insights into how lightweight models process information, highlighting the specific research gap that the proposed input versus output masking paradigm addresses.

Dataset Pruning and Curriculum Learning via Difficulty Scoring

The long-standing heuristic that expanding dataset size universally improves model capability has been systematically dismantled in the context of supervised fine-tuning. Recent empirical phenomena and theoretical analyses demonstrate that effective data selection must adapt dynamically to both the available data scale and the specific architectural capability of the target model.² A pivotal finding in recent literature is that preference data varies significantly in cognitive difficulty, and exposing a model to overly difficult examples actively hinders alignment by exceeding the model's parameter capacity.⁴ This underscores a critical relationship between data selection and model scale: a 0.5-billion parameter model simply cannot process the same complexity of training data as a 70-billion parameter model without suffering representational collapse.

To operationalize the concept of "difficulty," researchers have historically relied on perplexity-based filtering methodologies. A large perplexity score typically indicates that a data sample is difficult, rare, or anomalous relative to the model's pre-trained distribution.⁸ Early implementations utilized fixed n-gram language models to calculate perplexity across candidate datasets, typically selecting examples with intermediate values to form optimal training curriculums.² As LLMs evolved, these metrics became self-referential. Advancements in this space introduced the Instruction-Following Difficulty (IFD) metric, a self-guided method that quantifies the challenge of responding to a specific instruction without relying on external evaluator models or human annotation.⁹

The IFD score is mathematically defined as the ratio of the perplexity of the response conditioned on the instruction to the unconditional perplexity of the response.¹¹ A lower IFD score indicates that the instruction significantly aided the model in generating the response, whereas a higher score suggests a semantic disconnect or high inherent difficulty.¹¹ Further enhancements to this concept, such as the Token-Selective Hierarchical Data Selection (T-SHIRT) framework, identified critical limitations in sample-level evaluation.¹² T-SHIRT introduced the Selective-IFD (S-IFD) score, which isolates token-level informativeness, thereby preventing the erroneous selection of samples based on superficial lexical features rather than true semantic or structural quality.¹²

Furthermore, research into Adaptive Difficulty in reinforcement learning has demonstrated that

prioritizing questions of moderate difficulty yields the most informative learning signals, drastically reducing fine-tuning time while maintaining state-of-the-art performance benchmarks.¹ This principle of targeting the "median difficulty bracket" forms the theoretical justification for the data filtering mechanisms proposed in this experimental framework.

Token-Masking Mechanisms and Forgetting Mitigation

Token-masking has traditionally been associated with pre-training objectives, most notably Masked Language Modeling (MLM) utilized in BERT architectures, as well as visual input masking tasks like Masked Region Modeling (MRM).¹⁵ However, recent literature has repurposed masking as a highly effective tool for data curation, dataset pruning, and the active mitigation of catastrophic forgetting during the fine-tuning phase.

The introduction of Selective Token Masking (STM) represents a significant advancement in this specific domain. STM operates on the principle that fine-tuning an LLM on high-perplexity tokens correlates strongly with the degradation of a model's general, out-of-domain capabilities.¹⁷ By actively filtering out or masking these high-perplexity tokens during the supervised fine-tuning stage, STM blocks backpropagation from these unstable elements, thereby enabling what researchers term "low-perplexity training".¹⁸ Empirical evidence robustly demonstrates that STM not only improves target task performance but virtually eliminates the backward transfer of catastrophic forgetting, achieving capabilities comparable to models trained on highly curated, computationally expensive self-generated data.²⁰

Moreover, alternative masking mechanisms have been explored to enhance pure reasoning capabilities. For instance, "retrieved token masking" prevents unintended learning dynamics by excluding specific contextual elements from the reinforcement learning loss computation, yielding substantial improvements in complex question-answering benchmarks.²¹ Other innovative approaches, such as DiffuMask, utilize diffusion language models to predict token retention masks entirely in parallel, enabling efficient and scalable prompt compression by isolating hierarchical pruning behaviors.²²

The Research Gap: Differentiated Masking for Diagnostic Curation

While the existing literature heavily utilizes perplexity and token masking, it predominantly applies these concepts in a uniform, reactive, or protective manner. Methods like STM mask high-perplexity tokens in the target sequence specifically to protect the model's weights during backpropagation.¹⁹ Conversely, difficulty metrics like IFD compare full prompt conditioning against zero prompt conditioning to generate a passive difficulty score.¹¹

The specific, highly novel research gap this thesis fills is the active, differentiated use of selective token masking (Input vs. Output) as an upfront diagnostic tool to generate nuanced difficulty scores for data curation, rather than deploying it merely as a protective training mechanism.

Currently, no established framework systematically compares how input masking (perturbing the prompt to test generation stability) versus output masking (perturbing the target to test prediction certainty) differs in identifying the optimal curriculum for a lightweight model. Strategy A hypothesizes that semantic understanding is highly sensitive to input context; thus,

input masking will effectively isolate samples that require robust reasoning over fragile contextual cues. Strategy B hypothesizes that logical and mathematical reasoning relies on strict sequential certainty; thus, output masking will effectively isolate samples where the model's internal step-by-step derivation is unstable. By bridging token-masking mechanics with difficulty-based dataset pruning, this research introduces a highly distinct, computationally efficient paradigm for self-curated SFT alignment.

Mathematical Framework and Formal Definitions

To implement this comparative curation strategy with academic rigor, it is necessary to construct a formal mathematical framework that translates the conceptual strategies of Input Masking and Output Masking into quantifiable "difficulty scores." These scores must be mathematically sound, grounded in information theory, yet computationally inexpensive to calculate during a single forward pass of the Qwen-2.5-0.5B architecture.

1. Formalization of Baseline Probability Metrics

Let θ represent the trained parameters of the baseline language model. For a given dataset sample consisting of an input instruction sequence $x = (x_1, x_2, \dots, x_N)$ and a corresponding target response sequence $y = (y_1, y_2, \dots, y_T)$, the standard objective of causal language modeling is to map the conditional probability $P_\theta(y|x)$. The auto-regressive generation probability of the response given the input is defined by the standard chain rule of probability:

$$P_\theta(y|x) = \prod_{t=1}^T P_\theta(y_t | y_{<t}, x)$$

The standard cross-entropy loss (or negative log-likelihood) for the target sequence, which forms the basis of traditional perplexity metrics, is expressed as:

$$L_{CE}(x, y) = -\frac{1}{T} \sum_{t=1}^T \log P_\theta(y_t | y_{<t}, x)$$

While L_{CE} provides a baseline measure of how "surprised" the model is by the target sequence, it does not isolate the specific dependencies the model has on the input context versus its own internal sequential logic. To decouple these dependencies, the masking strategies must be mathematically formalized.

2. Strategy A: Input Masking Mechanics (D_{in})

Strategy A seeks to evaluate the model's generation stability by masking key informational tokens in the prompt x . The governing hypothesis is that if a sample is "easy" for the model, it relies heavily on a few key input tokens to trigger a strong internal pathway. If those specific tokens are masked, the target probability should collapse significantly. If the sample is inherently "difficult," the model's understanding is already fragmented, and masking input tokens will result in a lower relative variance in target probability.

Let $M_{in}(x)$ represent the masked input sequence, where a specific subset of highly informative tokens in x (e.g., specific nouns, definitive mathematical operators) are replaced by a generic `<[mask]>` token or functionally dropped from the attention mechanism. The conditional probability of the target sequence under input masking conditions is defined as:

$$P_{in}(y|M_{in}(x)) = \prod_{t=1}^T P_{\theta}(y_t|y_{<t}, M_{in}(x))$$

The Input Masking Difficulty Score, denoted as $D_{in}(x, y)$, is defined as the log-probability gap between the unmasked and masked conditions. To ensure computational efficiency (requiring only two forward passes without backpropagation), we utilize the average token-level log-probability difference, conceptually mirroring the Kullback-Leibler (KL) divergence between the two conditional states:

$$D_{in}(x, y) = \frac{1}{T} \sum_{t=1}^T \left(\log P_{\theta}(y_t|y_{<t}, x) - \log P_{\theta}(y_t|y_{<t}, M_{in}(x)) \right)$$

Mathematical Interpretation and Mechanistic Relevance: A high D_{in} indicates high generation instability; the model's ability to accurately predict the target y is highly dependent on the specific masked tokens present in x . Therefore, the instruction provides critical conditioning, and the model recognizes this dependence. A low D_{in} (approaching zero) indicates that the model either already predicts y purely from $y_{<t}$ (suggesting memorization or generic output), or the input x was fundamentally incomprehensible to the model.

regardless of the mask, signifying extreme difficulty that exceeds the model's capacity.⁴

3. Strategy B: Output Masking Mechanics (D_{out})

Strategy B evaluates prediction certainty by masking target tokens or answer sequences. This strategy is particularly relevant for the logical/mathematical data track, where each step in a Chain-of-Thought (CoT) sequence is rigorously dependent on the preceding step.

Let $M_{out}(y)$ represent a partially masked historical target sequence, where specific tokens in the target y are hidden during the evaluation of subsequent tokens. In a standard auto-regressive forward pass, the model attends to all $y_{<t}$ to predict y_t . Under Output Masking, we force the model to predict y_t while lacking access to key previous tokens y_k (where $k < t$).

Let $\tilde{y}_{<t}$ denote this selectively masked historical target context. The probability of predicting the full unmasked sequence under output masking conditions is evaluated as:

$$P_{out}(y|x) = \prod_{t=1}^T P_{\theta}(y_t|\tilde{y}_{<t}, x)$$

The Output Masking Difficulty Score, denoted as $D_{out}(x, y)$, is calculated as the variance or degradation in prediction certainty when historical target context is compromised:

$$D_{out}(x, y) = \frac{1}{T} \sum_{t=1}^T \left(\log P_{\theta}(y_t|y_{<t}, x) - \log P_{\theta}(y_t|\tilde{y}_{<t}, x) \right)$$

Mathematical Interpretation and Mechanistic Relevance:

A high D_{out} implies that the model possesses high sequential dependency; removing historical target tokens effectively breaks the logical chain, causing localized uncertainty to spike. This indicates a sample requiring rigorous step-by-step logic (ideal for mathematical reasoning tasks). A low D_{out} implies the target sequence is highly redundant or semantically fluid, where missing previous words does not prevent the prediction of future words (ideal for general conversational chat, but poor for strict logic).

4. Difficulty Assignment and Optimal Subset Selection

To utilize D_{in} and D_{out} effectively for dataset curation, we must map these raw divergence scores into a normalized "Difficulty Bracket" that dictates inclusion in the final fine-tuning set.

For a given raw dataset $\mathcal{D}$, we compute the requisite scores for all samples $(x^{(i)}, y^{(i)}) \in \mathcal{D}$. We then normalize the scores across the dataset using standard z-score normalization to account for sequence length variations and inherent baseline model biases:

$$\hat{D}_{in}^{(i)} = \frac{D_{in}^{(i)} - \mu(D_{in})}{\sigma(D_{in})}$$

$$\hat{D}_{out}^{(i)} = \frac{D_{out}^{(i)} - \mu(D_{out})}{\sigma(D_{out})}$$

Following the prevailing principle that overly difficult examples exceed model capacity and overly easy examples provide negligible gradient signal¹, the optimal SFT subset $\mathcal{D}_{optimal}$ is constructed by selecting samples that fall strictly within the median difficulty bracket.

Specifically, we apply a threshold parameter τ to filter samples whose normalized difficulty falls within the interquartile range (IQR):

$$\mathcal{D}_{StrategyA} = \left\{ (x^{(i)}, y^{(i)}) \in \mathcal{D} \mid -\tau \leq \hat{D}_{in}^{(i)} \leq \tau \right\}$$

$$\mathcal{D}_{StrategyB} = \left\{ (x^{(i)}, y^{(i)}) \in \mathcal{D} \mid -\tau \leq \hat{D}_{out}^{(i)} \leq \tau \right\}$$

This mathematical formulation is highly computationally cheap. It requires exactly three forward passes per sample during the curation phase: one standard unmasked pass, one with $M_{in}(x)$, and one with $M_{out}(y)$. No gradient updates, backpropagation, or external evaluator models are required during curation, making it highly suitable for evaluating tens of thousands of samples rapidly on a single consumer GPU prior to fine-tuning.

Experimental and Architectural Design

The transition from mathematical theory to empirical execution requires a meticulously defined experimental protocol. This section details the architectural specifics of the chosen baseline model, the curation of the source datasets, the procedural implementation of the masking

protocols, and the highly optimized fine-tuning recipe required to execute this methodology under stringent hardware constraints.

1. Baseline Architecture: Qwen-2.5-0.5B Specifications

The selection of Qwen-2.5-0.5B as the baseline model is central to the thesis premise. With 0.49 billion total parameters (and 0.36 billion non-embedding parameters), the model is small enough to train locally on consumer hardware while possessing an architecture sophisticated enough to yield measurable performance variances under masking conditions.⁶

The model architecture utilizes a standard Transformer configuration with RoPE (Rotary Positional Embeddings), SwiGLU activation functions, and RMSNorm for deep stabilization.⁶ It is constructed with 24 hidden layers, a hidden dimension size of 896, and employs Grouped Query Attention (GQA) with 14 query heads and 2 key-value heads.⁶ Notably, it supports a full context length of 32,768 tokens and demonstrates proficiency in both instruction following and structured data processing natively.²³

Because the parameter size is highly constrained, its representational capacity acts as a magnifying glass for dataset quality. Suboptimal, noisy, or overly complex data will quickly induce catastrophic forgetting of its pre-trained linguistic priors.¹⁷ Therefore, the differences between the datasets curated via Strategy A and Strategy B will be empirically pronounced and highly visible during evaluation.

2. Data Tracks and Open-Source Selection

To test the hypothesis that input masking optimizes semantic curation while output masking optimizes logical curation, the experiment requires two distinct, high-quality base datasets. These datasets must be clean, open-source, and appropriately sized for a 0.5B model to prevent prolonged processing bottlenecks.

Track Designation	Cognitive Area	Focus	Recommended Base Dataset	Justification for 0.5B Parameter Model
Track 1: Semantic	General Understanding & Linguistic Nuance		Alpaca-Cleaned or a curated subset of UltraChat	Provides diverse, conversational, and structurally varied semantic instructions. Cleaned versions prevent the model from learning grammatical noise, allowing the difficulty score to

			isolate semantic complexity. ²⁵ pure
Track 2: Logical	Mathematics, Reasoning & Deduction	GSM8K (Grade School Math) or a subset of DeepScaleR	GSM8K provides deterministic, step-by-step mathematical reasoning. ⁷ It is complex enough to challenge a 0.5B model but structured enough that output masking will accurately disrupt sequential certainty.

The initial candidate datasets will consist of approximately 20,000 samples each. During the curation phase, the mathematical framework will evaluate all 20,000 samples and filter them down to the median difficulty bracket, producing optimal 5,000-sample subsets for actual SFT.

3. Token Masking Protocol Implementation

The precise definition of *what* constitutes a masked token is critical to the integrity of the experiment. Random masking introduces unacceptable statistical noise; therefore, a deterministic, information-theoretic approach must be employed to guarantee reproducibility.

Protocol A: Input Token Masking Formulation

To mask the input x , we employ a localized, entropy-based selection criteria combined with basic Part-of-Speech (POS) tagging. Using a lightweight external library (e.g., SpaCy), we identify Nouns, Verbs, and Adjectives within the prompt context. We then selectively mask exactly 20% of these high-information tokens by replacing them with a distinct <|mask|> token. For the mathematical track, numerical values and distinct operational keywords (e.g., "calculate", "derive") are prioritized for masking. This directly targets the structural integrity of the prompt's instructions.

Protocol B: Output Token Masking Formulation

To mask the target output y , we utilize an iterative block-masking protocol. Because autoregressive models rely heavily on immediate historical context, masking isolated tokens is insufficient to disrupt sequence certainty. Instead, we mask sequential blocks (e.g., contiguous spans of 3-5 tokens) at calculated intervals across the target sequence during the evaluation forward pass. This forces the model to bridge larger contextual gaps, accurately reflecting the

structural stability of its internal reasoning chains.

4. Fine-Tuning Recipe for Single Consumer GPU

To guarantee the reproducibility of this methodology on a single consumer GPU (e.g., an NVIDIA RTX 3060 Ti with 8GB VRAM or a single A100 slice)²⁷, the fine-tuning framework strictly relies on Low-Rank Adaptation (LoRA) paired with 4-bit quantization (QLoRA).²⁸

The integration of 4-bit NormalFloat (NF4) quantization significantly reduces the memory footprint, enabling the training of the 0.5B model within a tight 4GB VRAM constraint, allowing the remaining VRAM to process larger batch sizes and extensive context windows efficiently.²⁷ The empirical parameters for the LoRA configuration are carefully calibrated to ensure that the parameter updates are expressive enough to learn the curated datasets without overfitting.²⁶

Hyperparameter	Value Configuration	Justification for Qwen-2.5-0.5B Architecture
Quantization	4-bit NF4	Reduces base model memory footprint, allowing for stable gradient updates and extensive context processing on consumer GPUs. ²⁷
LoRA Rank (r)	16	A rank of 16 provides sufficient dimensional capacity to learn both semantic nuances and logical structures without inducing parameter bloat. ²⁸
LoRA Alpha (α)	32	Standard scaling factor applied to maintain gradient magnitude parity with the chosen rank. ²⁸
Target Modules	q_proj, k_proj, v_proj, o_proj	Targeting all primary attention projections maximizes the adapter's ability to rewire the model's contextual routing effectively. ²⁶

Learning Rate	$2 \times$	Optimal for small-scale LoRA to prevent aggressive divergence while ensuring steady, reliable convergence across epochs. ²⁹
Batch Size	4 (Effective 16 via accumulation)	A micro-batch size of 4 with 4 gradient accumulation steps maximizes memory efficiency while stabilizing the overall loss curve. ²⁷
Epochs	3 to 5	Empirically sufficient for the lightweight model to internalize the 5,000-sample curated datasets without devolving into memorization. ³
Scheduler	Cosine Annealing	Ensures smooth convergence toward the end of training, preventing late-stage overfitting.

The experimental pipeline will result in the generation of four distinct model variants for comparative analysis:

1. **Model Baseline:** Zero-shot Qwen-2.5-0.5B (No tuning applied).
2. **Model Control:** Fine-tuned on a random 5,000 sample subset (No difficulty curation applied).
3. **Model Input-Masked:** Fine-tuned on the 5,000 samples curated via Strategy A (D_{in}).
4. **Model Output-Masked:** Fine-tuned on the 5,000 samples curated via Strategy B (D_{out}).

Evaluation Matrix and Comparative Analysis Framework

The evaluation matrix is fundamentally designed to definitively answer the core research question: Does input masking versus output masking yield measurably different alignment characteristics when applied to semantic versus logical tasks? Furthermore, the evaluation must meticulously quantify the extent to which each curation strategy induces or mitigates catastrophic forgetting of the model's pre-trained capabilities.

1. Benchmarking Semantic vs. Logical Alignment

To evaluate the fine-tuned models effectively, a diverse suite of standardized, zero-shot benchmarks will be employed. These benchmarks are purposefully selected to independently isolate semantic comprehension from strict deductive reasoning.

Semantic and Linguistic Assessment Benchmarks:

- **ARC-Challenge (ARC-C):** Evaluates the model's ability to answer complex, grade-school science questions requiring robust semantic comprehension and conceptual synthesis.³¹
- **HellaSwag (HS):** Tests the model's capacity to perform commonsense natural language inference and sentence completion. Success here strongly indicates that the model has successfully retained broad linguistic priors.³¹
- **MMLU (Massive Multitask Language Understanding):** Provides a comprehensive overview of the model's general knowledge retention and instruction-following capability across highly diverse subjects.³¹

Logical and Mathematical Assessment Benchmarks:

- **GSM8K Test Set:** Evaluates the model's deterministic, multi-step arithmetic reasoning. The accuracy of the final generated answer depends entirely on the stability of the intermediate logical steps.⁷
- **MATH Dataset:** A significantly more rigorous assessment of mathematical problem-solving, requiring advanced algebraic and geometric derivation, testing the absolute limits of the 0.5B parameter capacity.⁷

The comparative analysis will scrutinize the differential performance matrix resulting from these evaluations. If the central hypothesis holds true, the Input-Masked Model should demonstrate superior performance on ARC-C and HellaSwag, as its dataset was optimized for stable contextual mapping. Conversely, the Output-Masked Model should demonstrate superior performance on GSM8K and MATH, as its dataset was specifically optimized to enforce internal sequential rigor.

2. Post-Tuning Robustness and Forgetting Metrics

In addition to evaluating pure performance benchmarks, the framework must quantify the overall architectural stability of the models utilizing the Task Improvement (TI) and Backward Transfer (BWT) metrics established heavily in the Selective Token Masking literature.¹⁸

Catastrophic forgetting remains a primary, persistent risk in supervised fine-tuning, especially in models with less than one billion parameters.¹⁷ If the proposed difficulty curation strategies successfully isolate the optimal learning threshold without exceeding capacity, the resulting models should exhibit a highly positive Task Improvement on their target domains while maintaining a near-zero or minimal negative Backward Transfer on non-target domains.²⁰

By calculating the baseline perplexity of the fine-tuned models on a held-out corpus of raw text, the evaluation will accurately determine if the selective masking strategies inadvertently curated data distributions that warped the model's foundational linguistic geometry. The ability to enhance specific domain performance while concurrently minimizing out-of-domain degradation provides the ultimate empirical validation for the difficulty assignment

methodology.¹⁸

Zero-Risk Academic Framing and Contingency Analysis

A fundamental requirement of a rigorous academic Master's thesis is that the intrinsic value of the research remains absolute, irrespective of the final empirical outcomes. The methodological design inherently generates epistemological value. The "Zero-Risk Academic Framing" dictates exactly how the abstract, discussion, and conclusions must be positioned to ensure the research is highly publishable regardless of whether the central hypothesis is empirically validated, explicitly refuted, or yields null results.

1. Epistemological Value Independent of Empirical Triumph

The primary risk inherent in this thesis is that neither Strategy A (Input Masking) nor Strategy B (Output Masking) significantly outperforms the standard Control baseline model. In a traditional academic framing, this would be viewed as a failure of the hypothesis. However, under the zero-risk framework, a null or negative result is aggressively framed as a critical diagnostic discovery concerning the architectural limitations of sub-1-billion parameter models. If both masking strategies fail to yield superior alignment, the thesis will pivot to an in-depth analysis of representational fragility. The conclusion will argue that in highly compact architectures (such as the 0.49B parameter Qwen-2.5), the internal attention mechanisms do not distinctly decouple prompt context from sequential generation context. Therefore, perturbing the input and perturbing the output degrade the conditional probability field uniformly, rendering targeted token-masking an ineffective tool for difficulty-based curriculum learning at this specific scale. This establishes a highly valuable, publishable boundary condition: it proves that advanced masking-based curation techniques, which succeed robustly in larger 7B+ models³, possess a strict minimum parameter threshold to be effective.

Conversely, if one strategy severely degrades the model (e.g., Output Masking causes a complete collapse in GSM8K mathematical performance), this negative result provides profound insight into how small models learn logic. It would heavily suggest that 0.5B models do not learn "reasoning" as a robust, internal sequence of logical steps, but rather rely entirely on shallow heuristic mapping from the prompt context. The failure of the strategy, therefore, perfectly illuminates the mechanistic reality of the lightweight model.

2. Architectural Insights and Methodological Contributions

Regardless of the final benchmark scores, the development of the mathematical framework itself constitutes a primary academic contribution. The formalization of D_{in} and D_{out} as computationally cheap, forward-pass-only difficulty metrics provides a highly novel tool for the open-source NLP community. By demonstrating how to construct these sophisticated metrics using standard cross-entropy and probability outputs, the thesis contributes a reusable methodology for self-curation that entirely bypasses the need for massive, API-driven evaluator models like GPT-4, aligning with the industry push toward decentralized AI

development.⁹

Furthermore, the extensive documentation of the single-GPU fine-tuning recipe, including the exact behavioral response of the Qwen-2.5-0.5B architecture to 4-bit QLoRA adaptation under heavily filtered datasets, provides critical benchmarking data for researchers operating under severe hardware constraints. The thesis will position the comparative analysis not merely as a contest between two filtering algorithms, but as a comprehensive, mechanistic autopsy of how lightweight attention mechanisms respond to forced informational deficits during the critical curation phase.

By meticulously decoupling the empirical success of the fine-tuned models from the analytical success of the investigation, the thesis guarantees that its architectural insights, mathematical definitions, and evaluations of parameter boundaries remain a permanent, valuable addition to the literature on low-resource language model optimization.

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